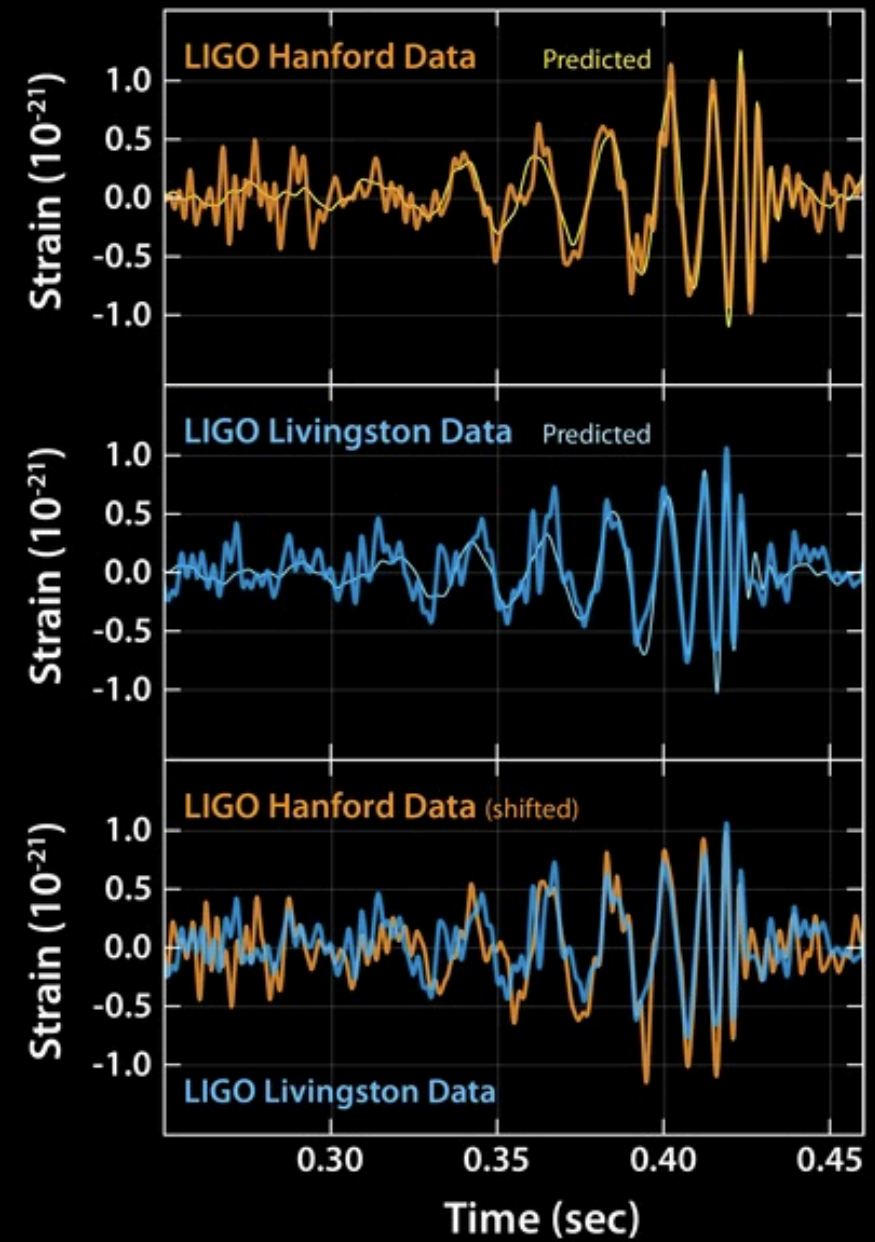
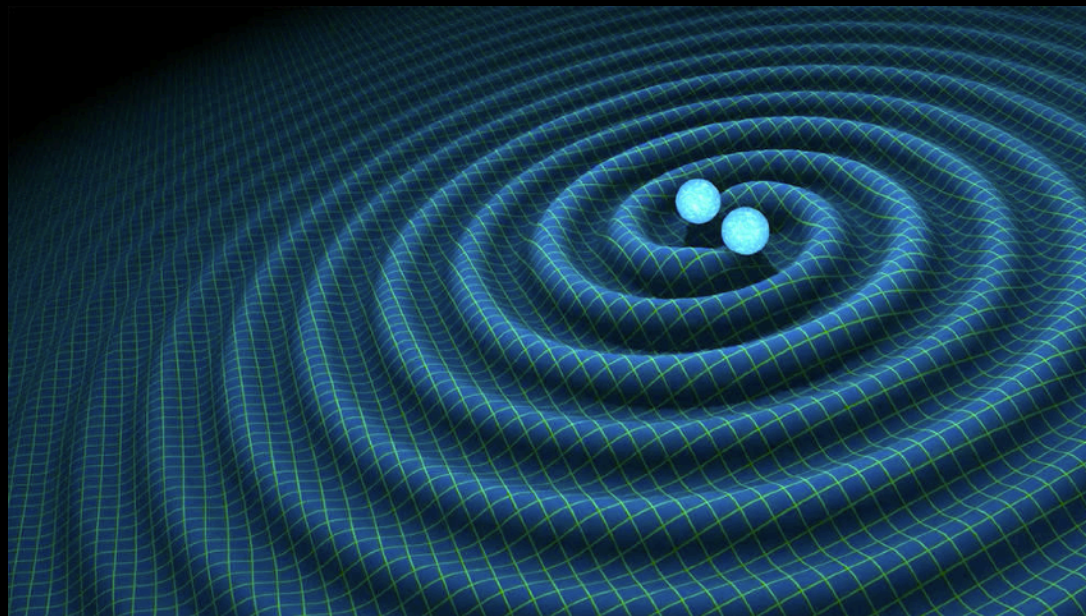
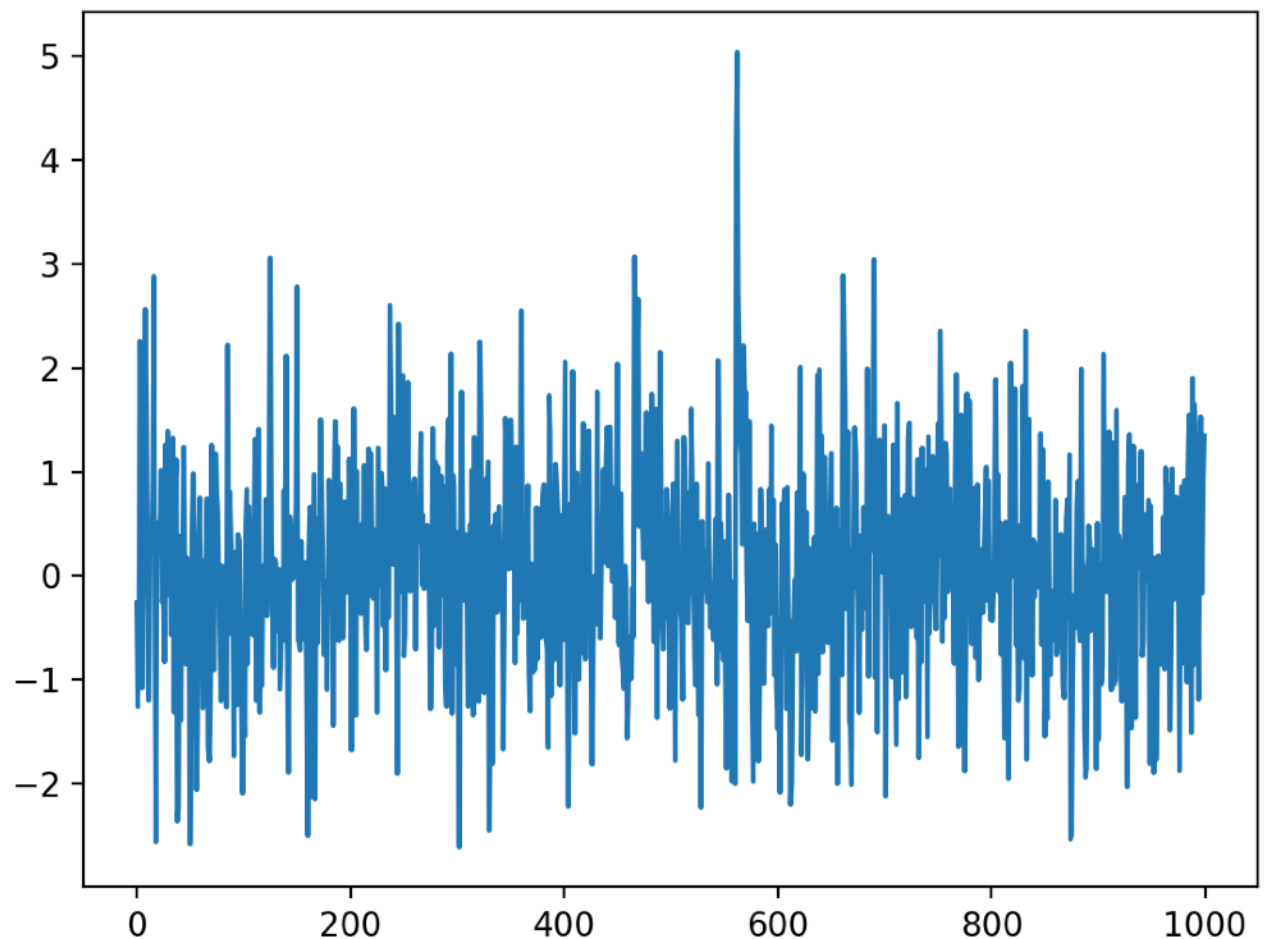


Matched Filters/Ligo



Searching for Signals

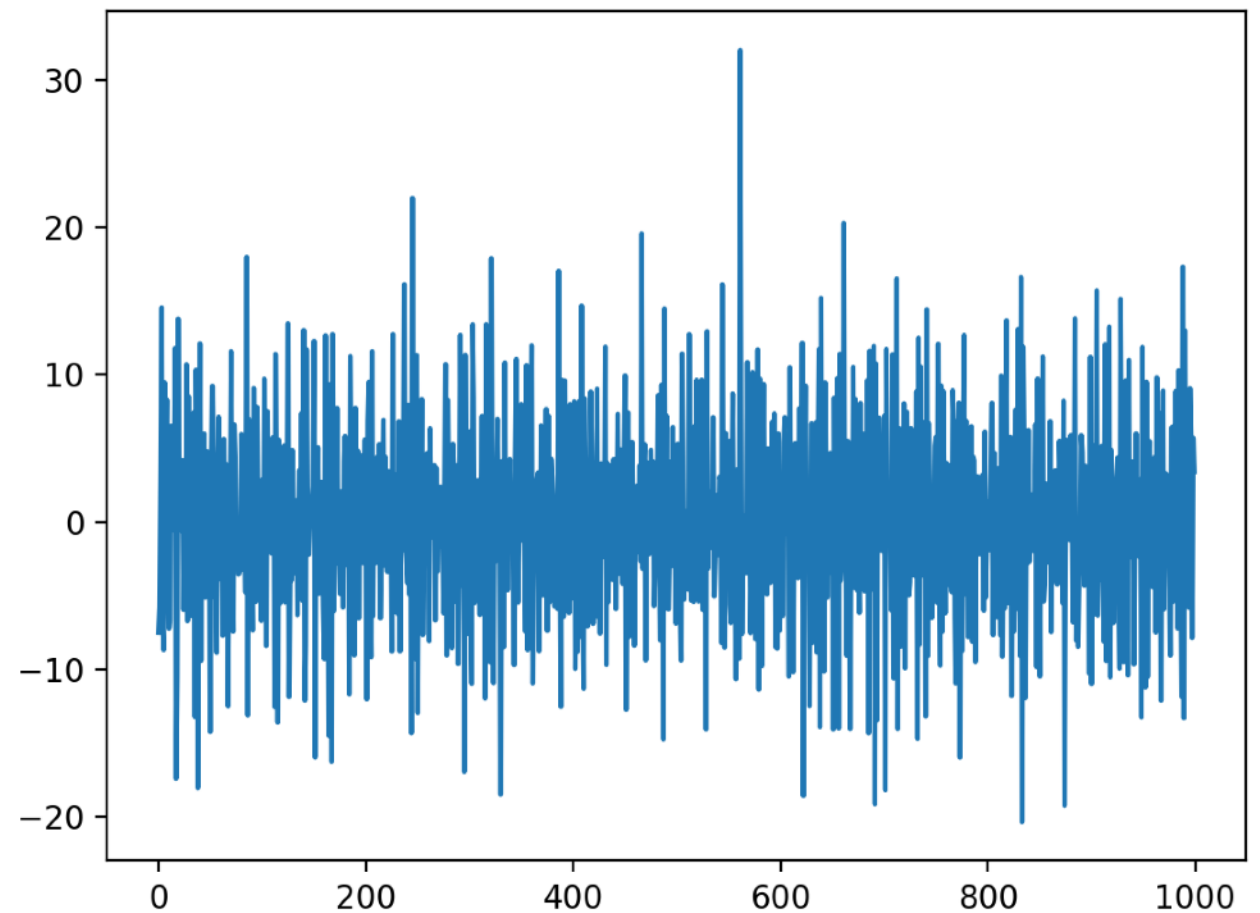
- Simulated data - particle detector.
- Signal decays exponentially when hit by particle.
- Noise is white.
- Where/how energetic were the particles that hit?



Simulated data of exponential-decay detector hit by particles.

Deconvolution

- Simplest attempt is to deconvolve observed data using exponential response.
- How might this work or not work? Why?



Deconvolved version of previous

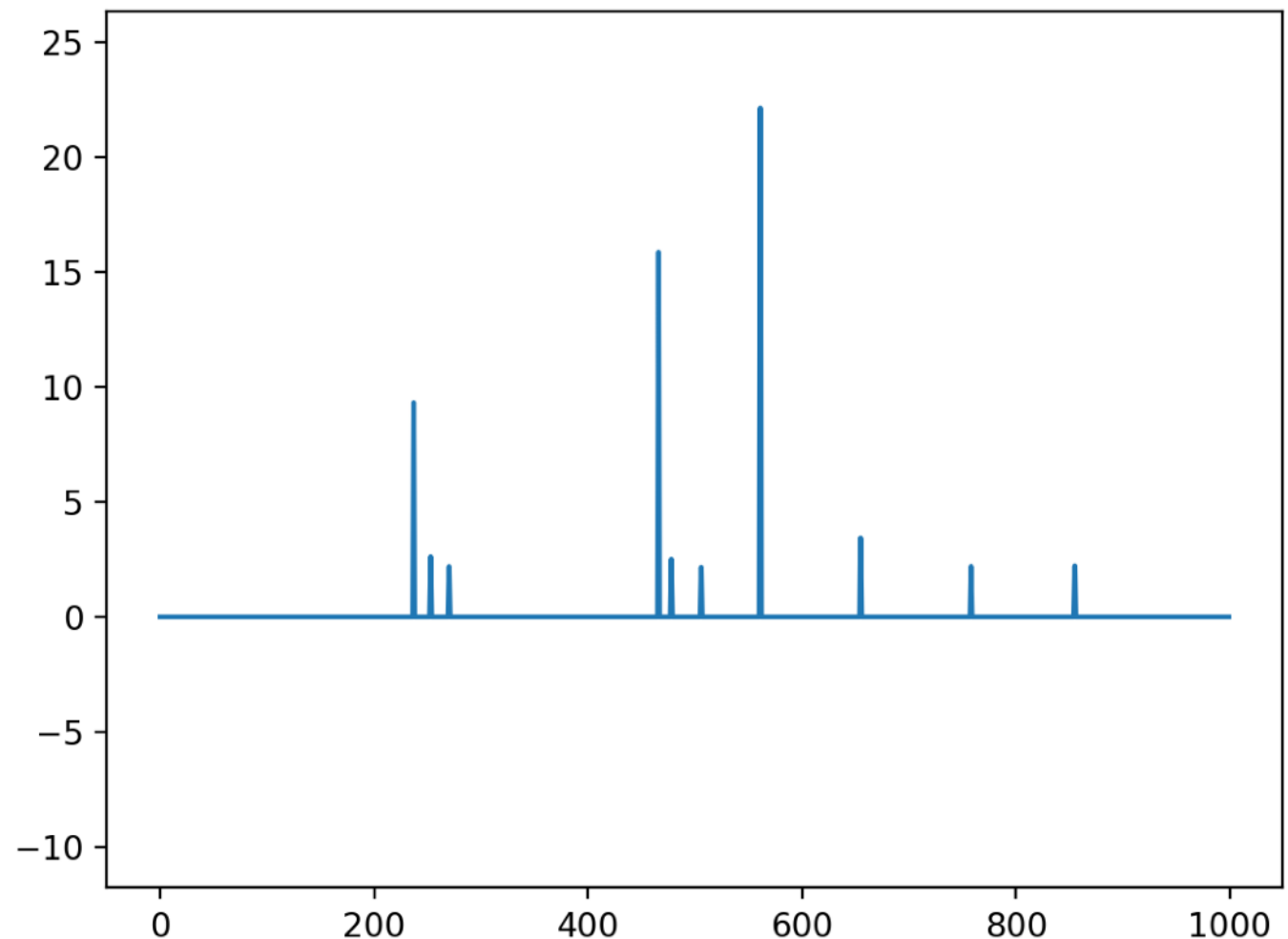
Matched Filter

- We want to search for a signal in data. We don't know where it will be. How do we find it?
- Best fit amplitude for 1-D template A is $A^T N^{-1} d / A^T N^{-1} A$
- We can search many possible locations of template with matched filter, replacing top by correlation of A with $N^{-1} d$ (or $N^{-1} A$ with d) if noise is stationary
- Alternatively, could take correlation of $N^{-1/2} A$ with $N^{-1/2} d$. What would the noise in $N^{-1/2} d$ look like?

NB - relevant operation is cross-correlation, not convolution

MF Output

- Output of MF, vs. true input signal.
- In this case, SNR of MF is twice that of deconvolution.
- MF *multiplies* by noise, deconv. *divides*. MF stays stable in presence of zeros in noise.

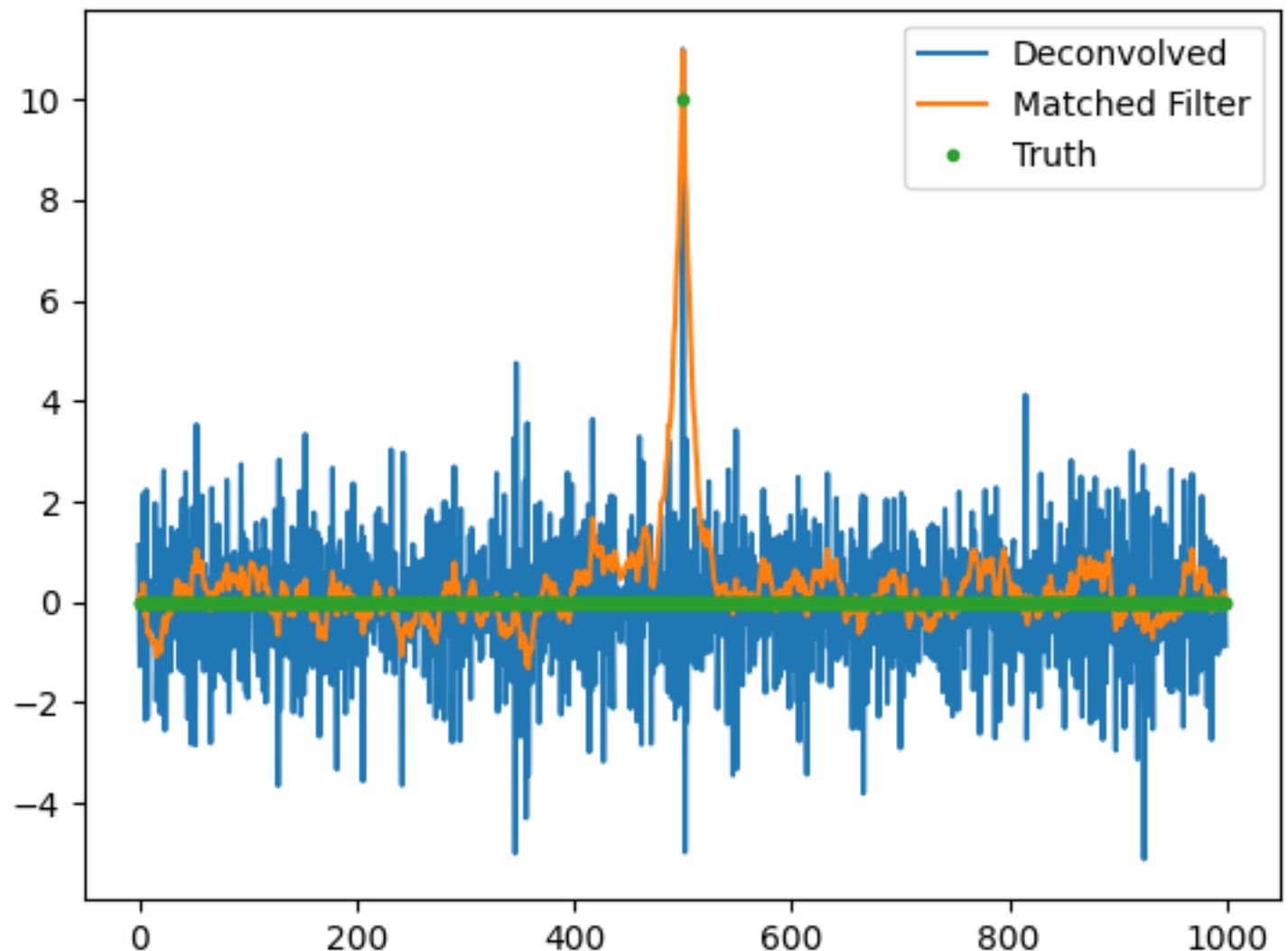


MF vs. Deconv

- Matched filter expects exactly one signal in the data.
- Deconvolution solves for signal at *every* data point.
- If your signals overlap, assumptions for MF are wrong. If they don't, deconv is higher noise.
- Use the right tool for the right job!

MF vs. Deconv Graph

- MF lower noise, but errors more correlated.
- Deconv higher noise, uncorrelated errors.
- Signals closer than MF correlation will interfere.

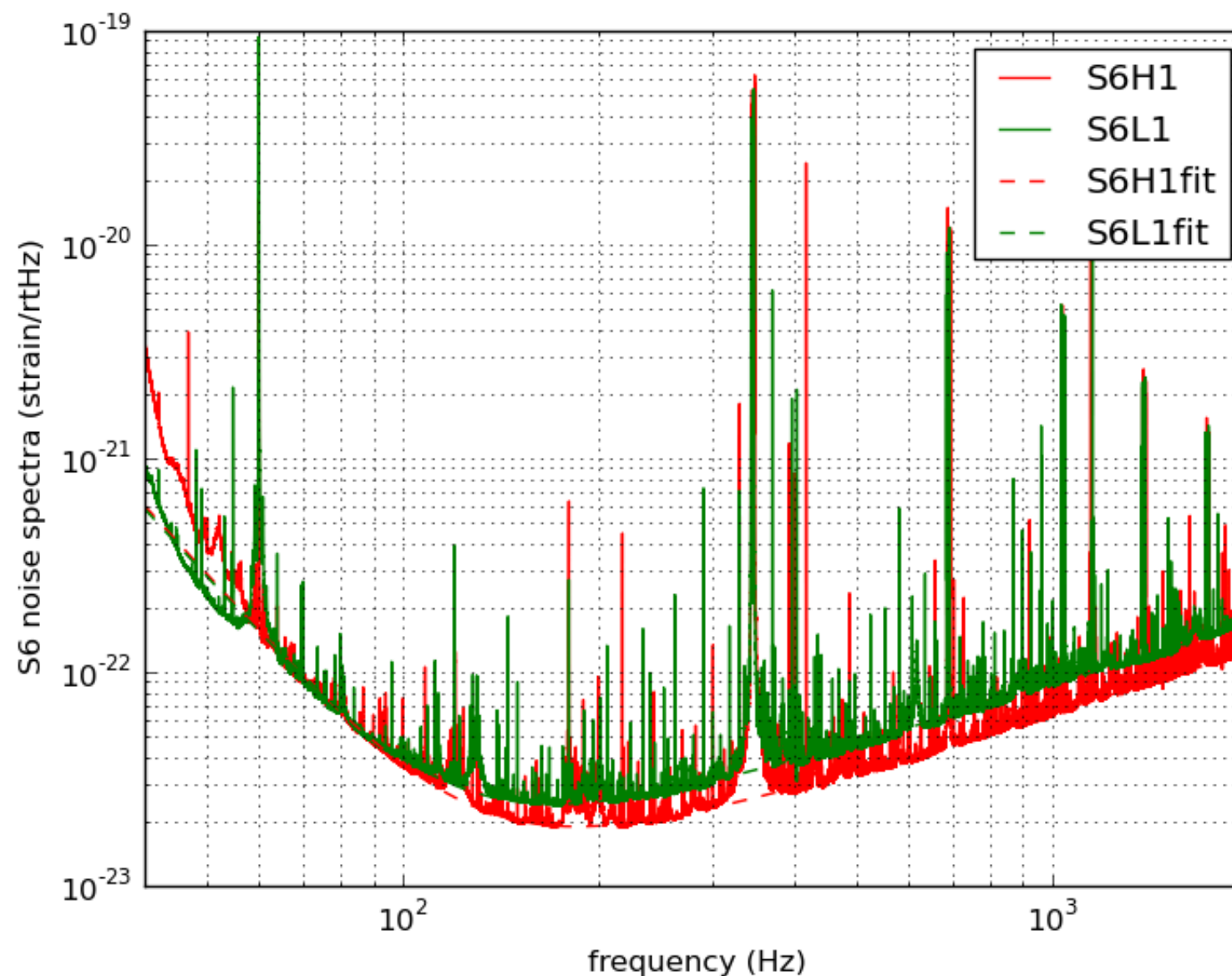


Output of mf_deconv_snr.py

LIGO Data

- https://github.com/losc-tutorial/LOSC_Event_tutorial
- alternatively, relevant files in ligo directory on course github (but losc-tutorial has more)
- `simple_read_ligo.py` will read for you (once you have h5py installed and working)

First, what should we see for noise?



Power Spectrum Description

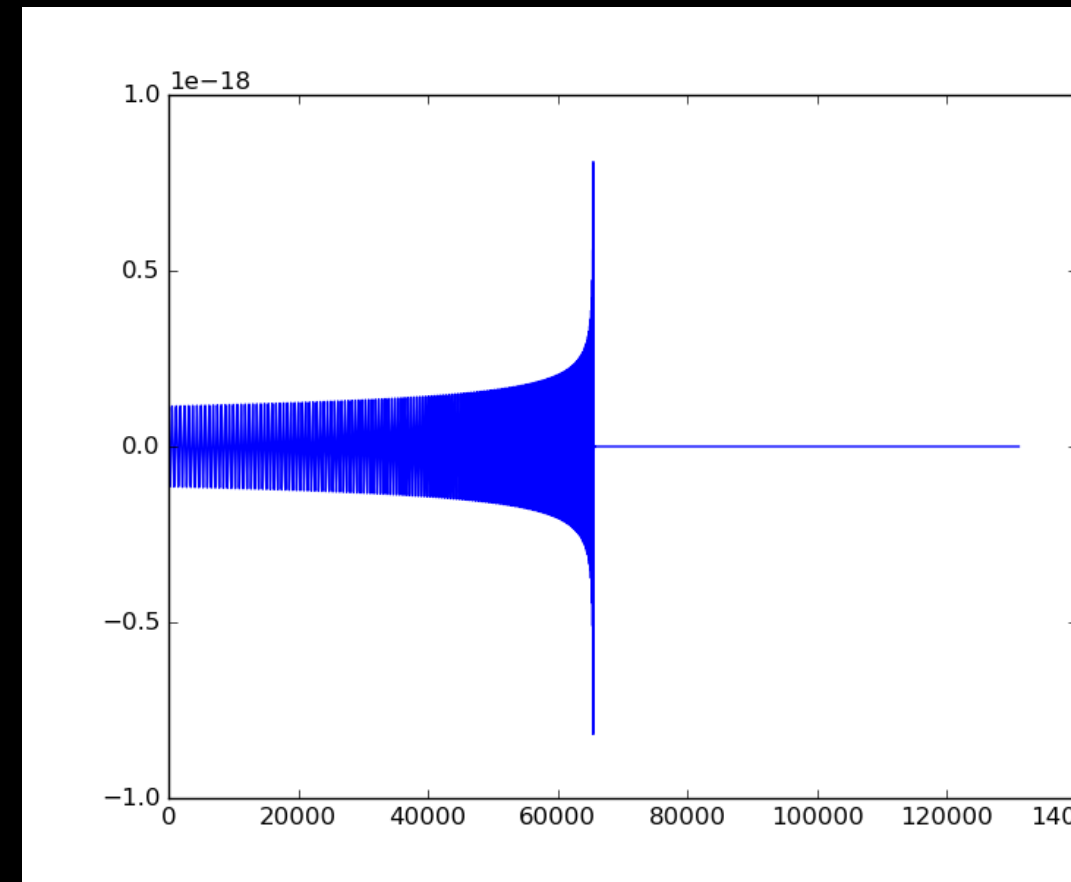
- χ^2 is the same in real or Fourier space
- Modes are uncorrelated in Fourier space. We'll describe noise there.
- SNR^2/mode is set by $(\text{template FT})^2/\text{noise PS}$
- Noise PS is just FT of correlation function

Noise in Fourier Space

- Let F be standard FT/ $\sqrt{n}$. Then $F^{-1} = F^\dagger$.
- $\chi^2 = r^T F^\dagger F N^{-1} F^\dagger F r$ where $r = d - A(m)$.
- Let $\tilde{r} = Fr$ and $\tilde{N} = F N F^\dagger$. We know $\tilde{N}$ is diagonal, so estimate from $\tilde{r}$. If noise $\gg$ signal, then $r \sim d$, so estimate from FT of data.
- Rewrite $\chi^2 = r F^\dagger \tilde{N} F r$. Then we know $N = F^\dagger \tilde{N} F$.
- Matched filter: We need $A^T N^{-1} d$ at every offset, so can correlated A against $N^{-1} d$ or d against $N^{-1} A$. We apply N^{-1} by taking FFT, dividing by (diagonal) $\tilde{N}$, and taking inverse FFT.

Pre-whitening

- One sometimes useful trick is to pre-whiten.
- Template is at right. Where does our SNR come from?
- Can't say because depends on details of N .
- Pre-whitening turns N into I , so can be obvious what's happening.



Pre-whitening

- Re-write least-squares solution: $(N^{-1/2}A)^T(N^{-1/2}A)m=(N^{-1/2}A)^T(N^{-1/2}d)$
- Let $A'=N^{-1/2}A$, $d'=N^{-1/2}d$, and get $A'^TA'm=A'^Td'$. Same as before, but now noise is identity. Hence, pre-whitening.
- $N^{-1/2}$ can be done by dividing by square root of noise.
- Look at pre-whitened template $(N^{-1/2}A)$ to see where you expect useful signal.
- Usual choice for $N^{-1/2}$ is $V\lambda^{-1/2}V^T$.

Fourier Interpretation

- Noise model has same total variance independent of correlation length.
- Looking at FT, long length packs noise power into many long wavelengths. Template has more power on high-frequency scales (good SNR)
- Short length spreads out power over many many modes, dropping average noise power. Template well above noise on large scales (good SNR).
- Intermediate packs all its noise into same scales as template. Never have good SNR.

When your noise looks like your signal,
you're going to have a bad day...

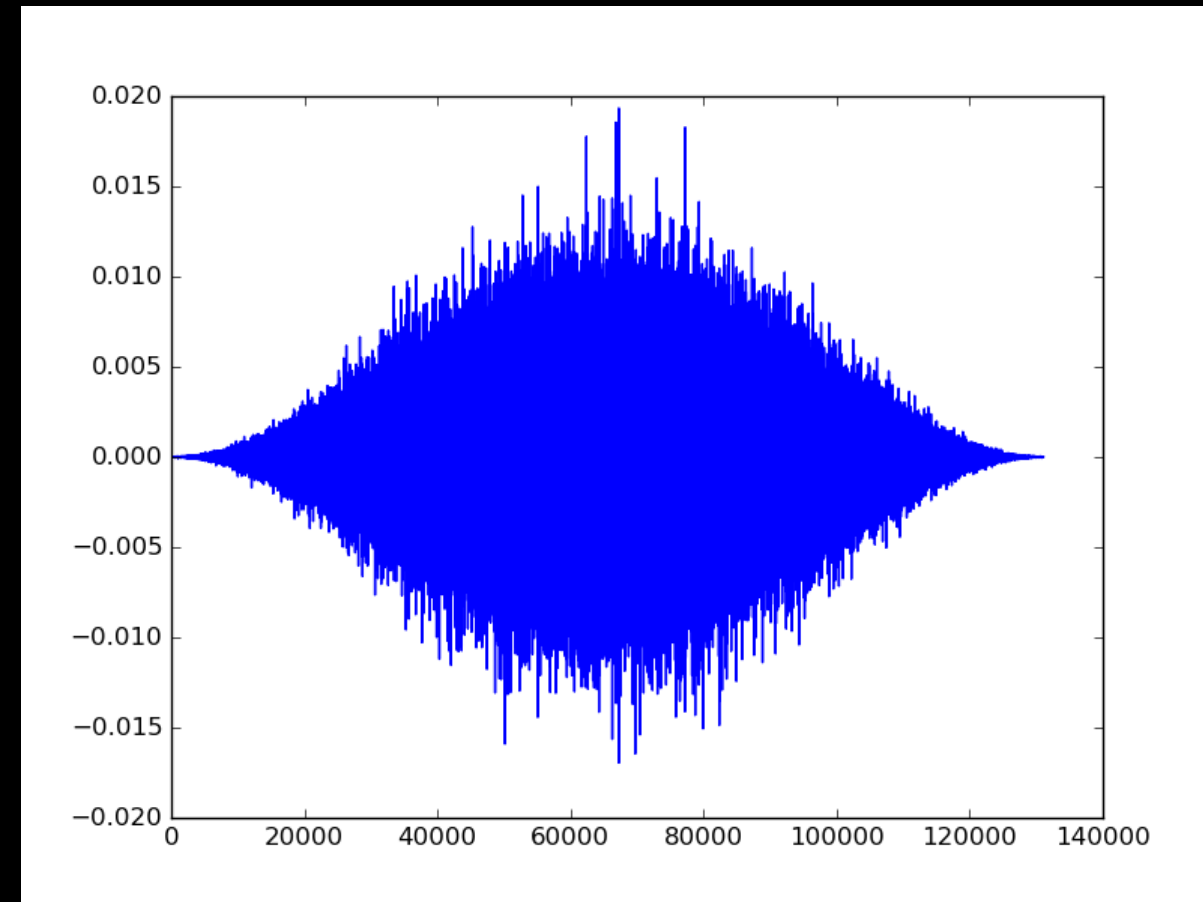
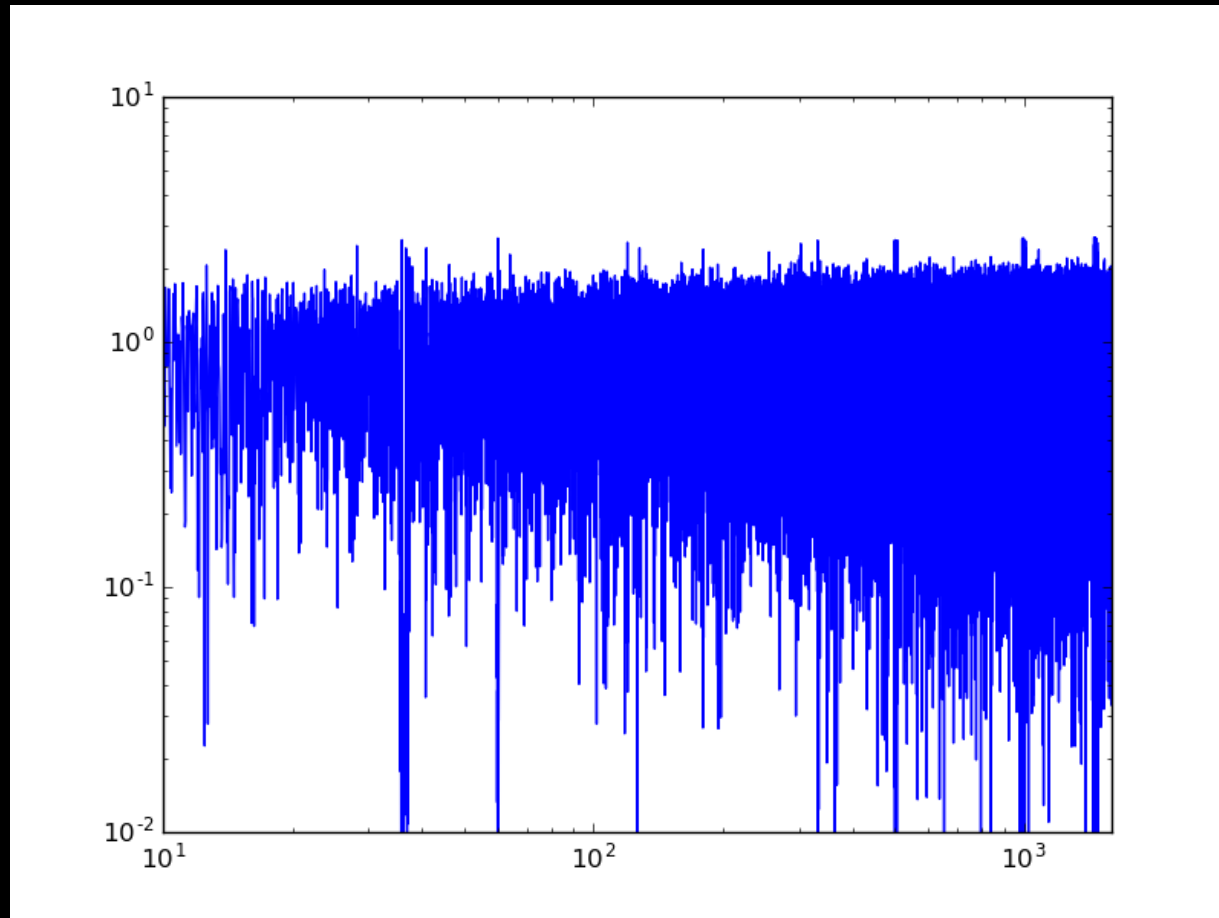
How Should We Estimate Noise?

- Windowing key to avoiding FFT ringing
- smooths out spectral features
- Noise large per mode in FT, so we have to average
- What are your thoughts on averaging?

Smoothing PS

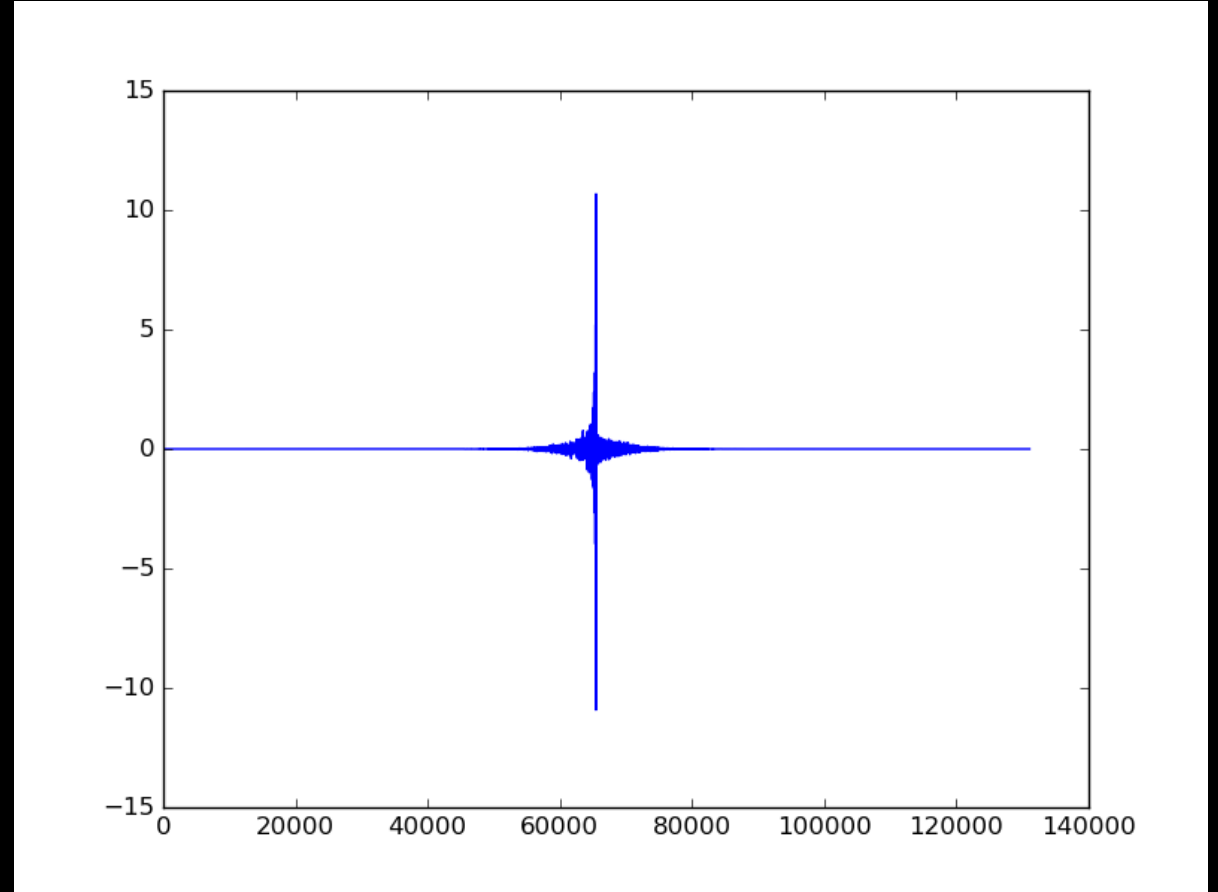
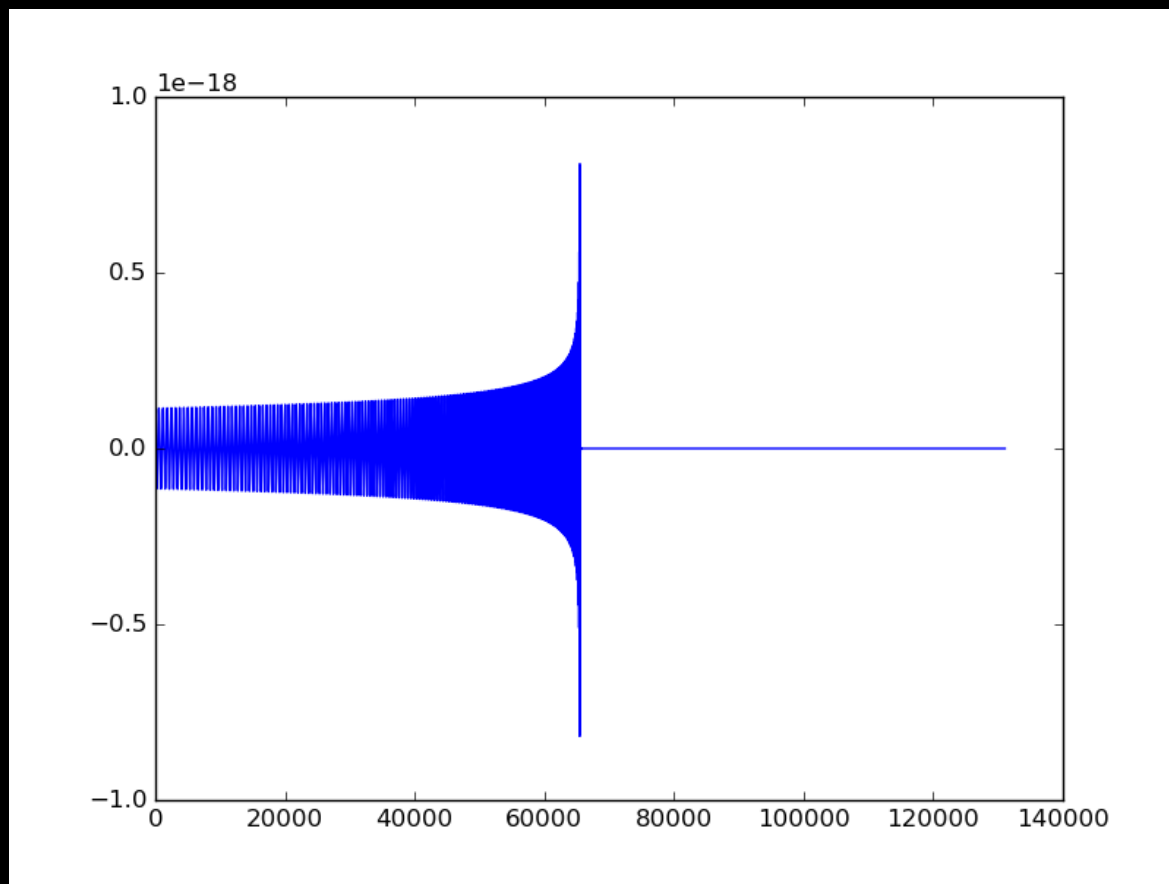
- Take $|FT|^2$, which is an estimate
- Smooth by convolving with an extended function.
- Thoughts on the function?

Pre-Whitened Data from Smoothed PS



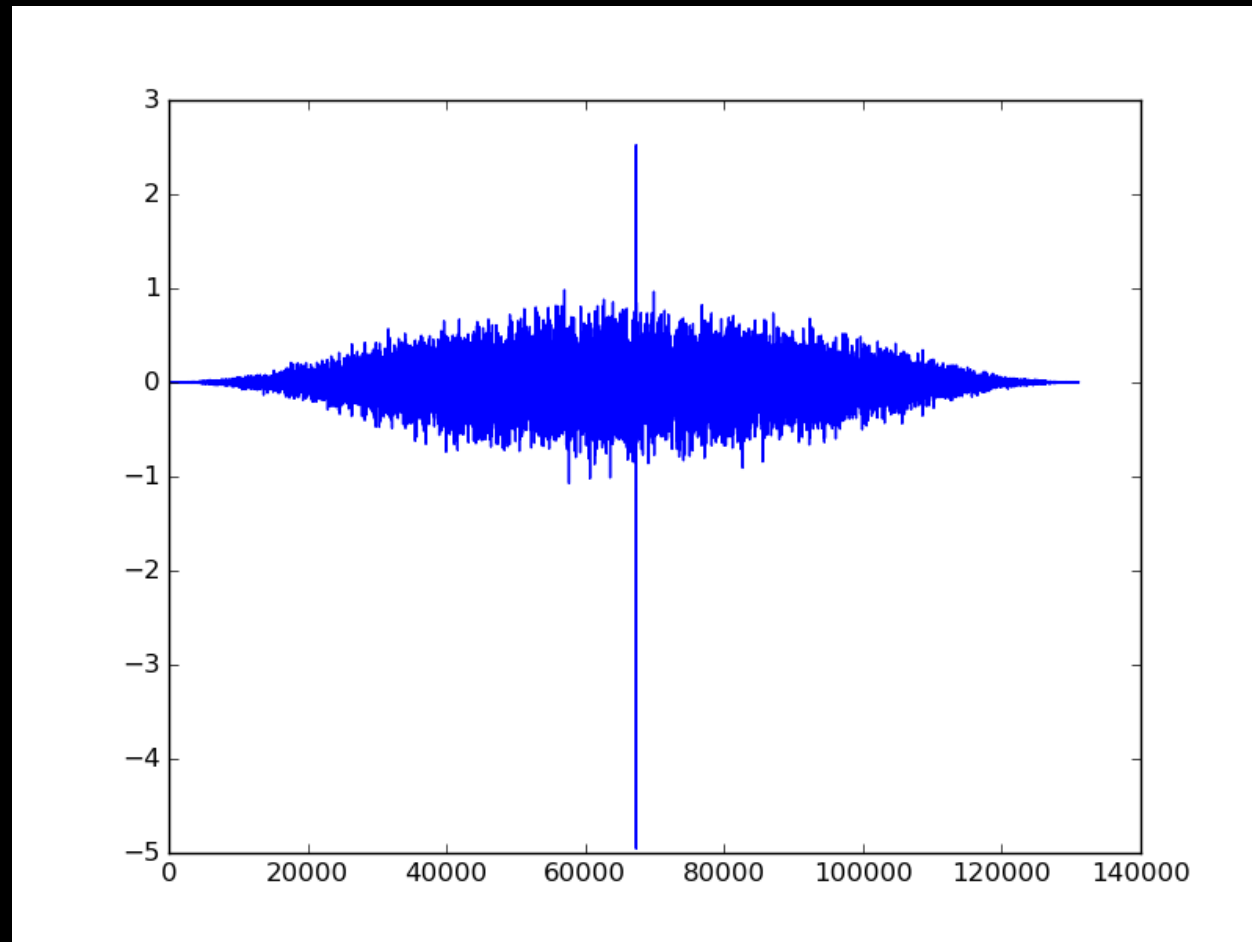
Left: Whitened FT of data. Looks not crazy. What are little nubbins sticking up?
Right: whitened data. Window shape is pretty obvious.

Pre-whitened template



- For this event, template is not small at start of data. Will this be a problem?
- Can look at pre-whitened version of template to get an idea.

Can Use for MF now



FFT Shift of matched filter output. We found a GW!

Averaging PS

- Break PS up into small chunks so we have many
- Take the FT of each chunk
- Add the FT²s together.
- How do we apply this (short) PS to original data?
- Qualitatively, how do we relate this PS estimate to smoothed one?

More Windowing

- Usual windows taper every sample.
- Reduces power in a way we probably aren't happy with
- How could we modify window to make this less of an issue?
- Let's try this on data...

Normalizations

- Properly normalizing noise can require care. I usually check with white noise.
- N^{-1} for white noise with $\sigma=1$ should be identity.
- Variance of FT is sum over data = $\sigma^2 N_{\text{data}}$. In Fourier space, we want $N(k) = \text{Var}(F(k)) / N_{\text{data}}$. Not just $\text{Var}(F(k))$
- Window function: $\text{Var}(F)$ for white = $\sum (\sigma^2 W^2)$.
- If you want the real-space variance to be correct where $W \sim 1$, you'll need to use $\sum (W^2)$ as your normalization.